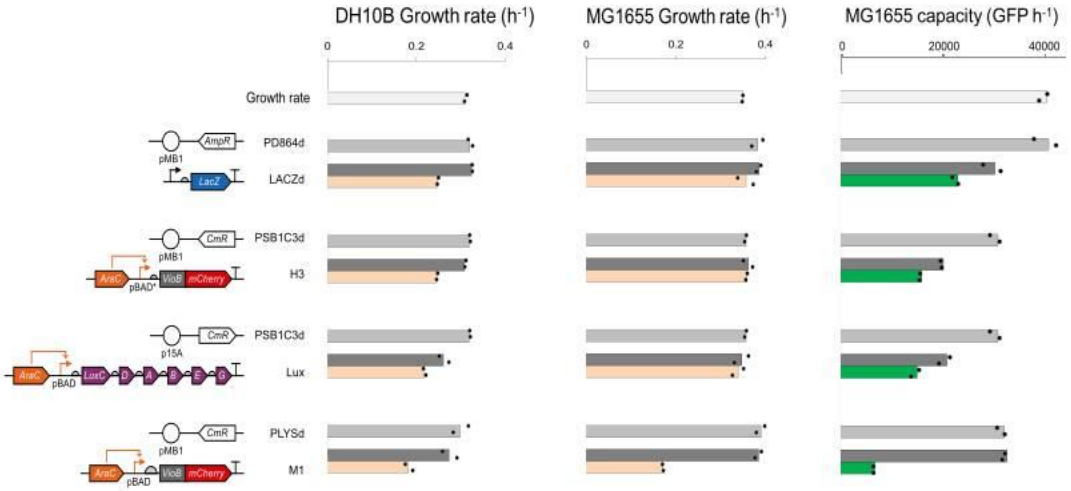
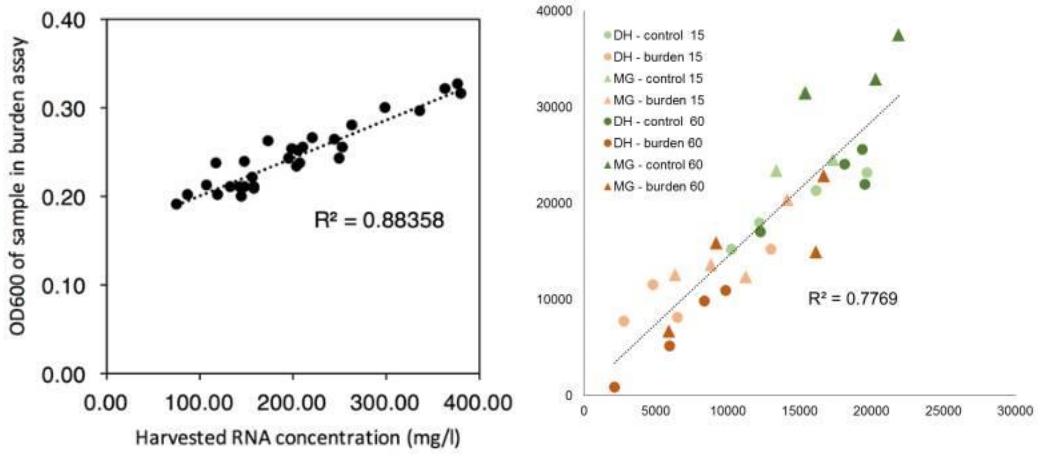


Supplementary Figure 1

Burden assay for cells transformed with the synthetic constructs 1 h after induction of gene expression.

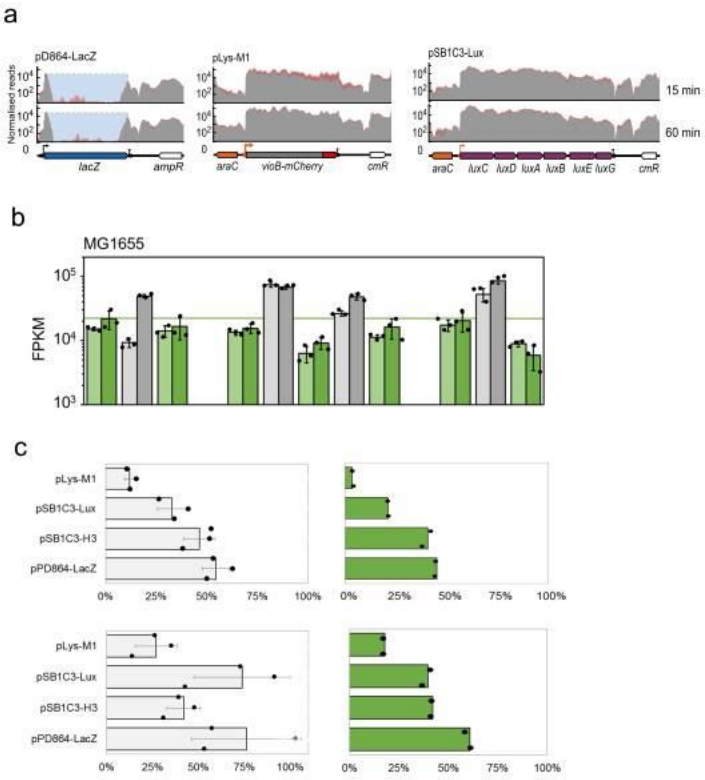
Left and middle panels show the growth rate in DH10B and MG1655, respectively. Grey and pink bars represent uninduced and induced samples, respectively. Right panel shows GFP production rate in MG1655. Grey and green bars represent uninduced and induced samples, respectively. Darker grey bars represent the uninduced samples; lighter grey bars represent the empty plasmids while the lighter top bars represent the growth rate in the untransformed strains. Dots show single repeats while bars show the mean value for two independent repeats.



Supplementary Figure 2

Correlations between RNA-seq data and the *in vivo* burden assay.

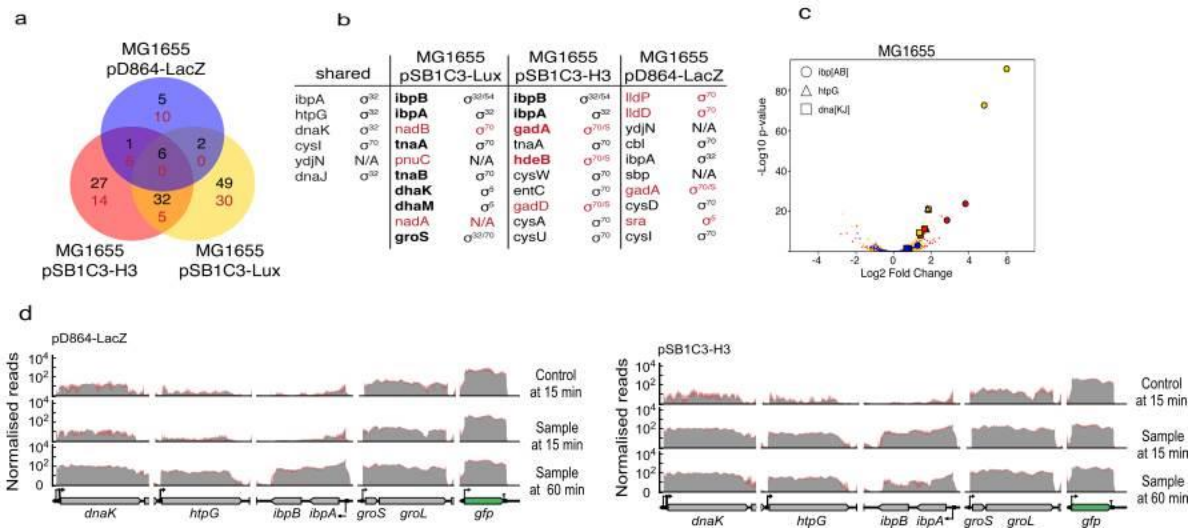
Left panel shows the relationship between the total RNA harvested during RNAseq sample prep and the total cells grown for each sample in the plate-based burden assay done in parallel. Correlation is given as R^2 - the square of the Pearson product moment correlation coefficient. Cell concentration per well was determined by OD600 measurement at each time point (15 min and 60 min post-induction). RNA concentrations from extractions into the same volume were determined by spectrographic measurement with Nanodrop. Data represent values from three independent repeats. Right panel shows the relationship between RNA-seq GFP FPKM (x-axis) and *in vivo* GFP production rate per cell (y-axis). Correlation is given as R^2 - the square of the Pearson product moment correlation coefficient. Data are from 15 and 60 min post-induction. Control samples are the strains (DH = DH10B, MG = MG1655) with the three empty plasmids (pSB1C3, pLys and pD864). Burden samples are the strains with the four synthetic constructs (pSB1C3-Lux, pSB1C3-H3, pLys-M1 and pD864-LacZ). Data show the mean values for each condition.



Supplementary Figure 3

Burden imposed by the four synthetic constructs.

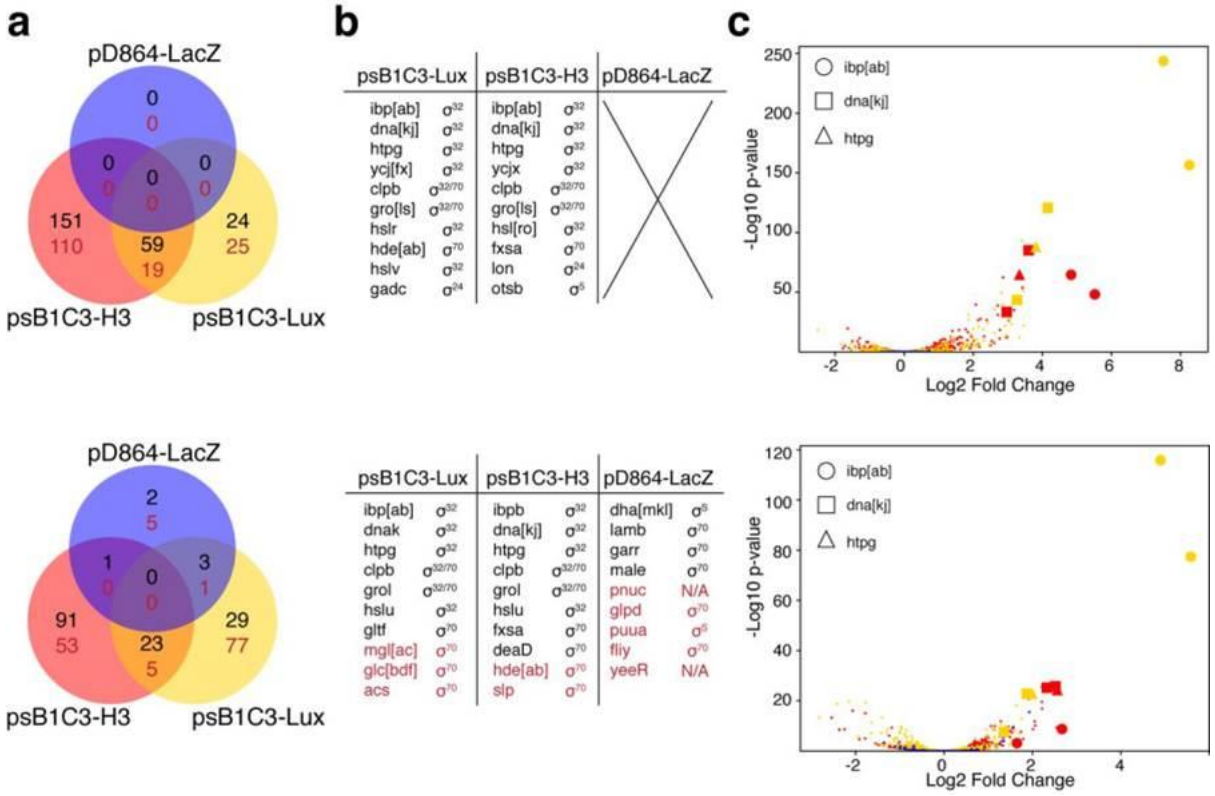
(a) Transcription profiles of the synthetic constructs pD864-LacZ, pLys-M1 and pSB1C3-Lux in DH10B cells 15 (top) and 60 (bottom) min post-induction**.** Traces show the normalized reads mapping to the plasmid constructs. Solid grey regions show average, and transparent red regions span from minimum to maximum of transcription profiles generated from three biological replicates. Genetic designs below each profile are drawn using SBOLv notation and show the precise location of each genetic part. For pD864-LacZ, only reads uniquely mapping to the plasmid *lacZ* gene (and not the chromosomal copy) are shown. These are used to estimate the level of plasmid *lacZ* expression (blue shaded regions, see Online Methods section). **(b)** Transcriptional induction of the four constructs represented by mean FPKM values in MG1655 at 1h post induction. Green bars are data for the calculated GFP FPKM and colour intensity relates to the time point; grey bars are data for the FPKM corresponding to the construct gene; * denotes empty plasmid controls. Dots and bars show individual measurements and their mean, respectively. Error bars show the standard deviation of three independent repeats. **(c)** Percentage of available capacity 1h post induction. In the left panels, plots show GFP mRNA FPKM in construct samples compared to the pD864 control. In the right panels, plots show the same comparison but using GFP fluorescence production rate data from the plate-based burden assay (Figure 1c). Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats.



Supplementary Figure 4

Transcriptome data analysis.

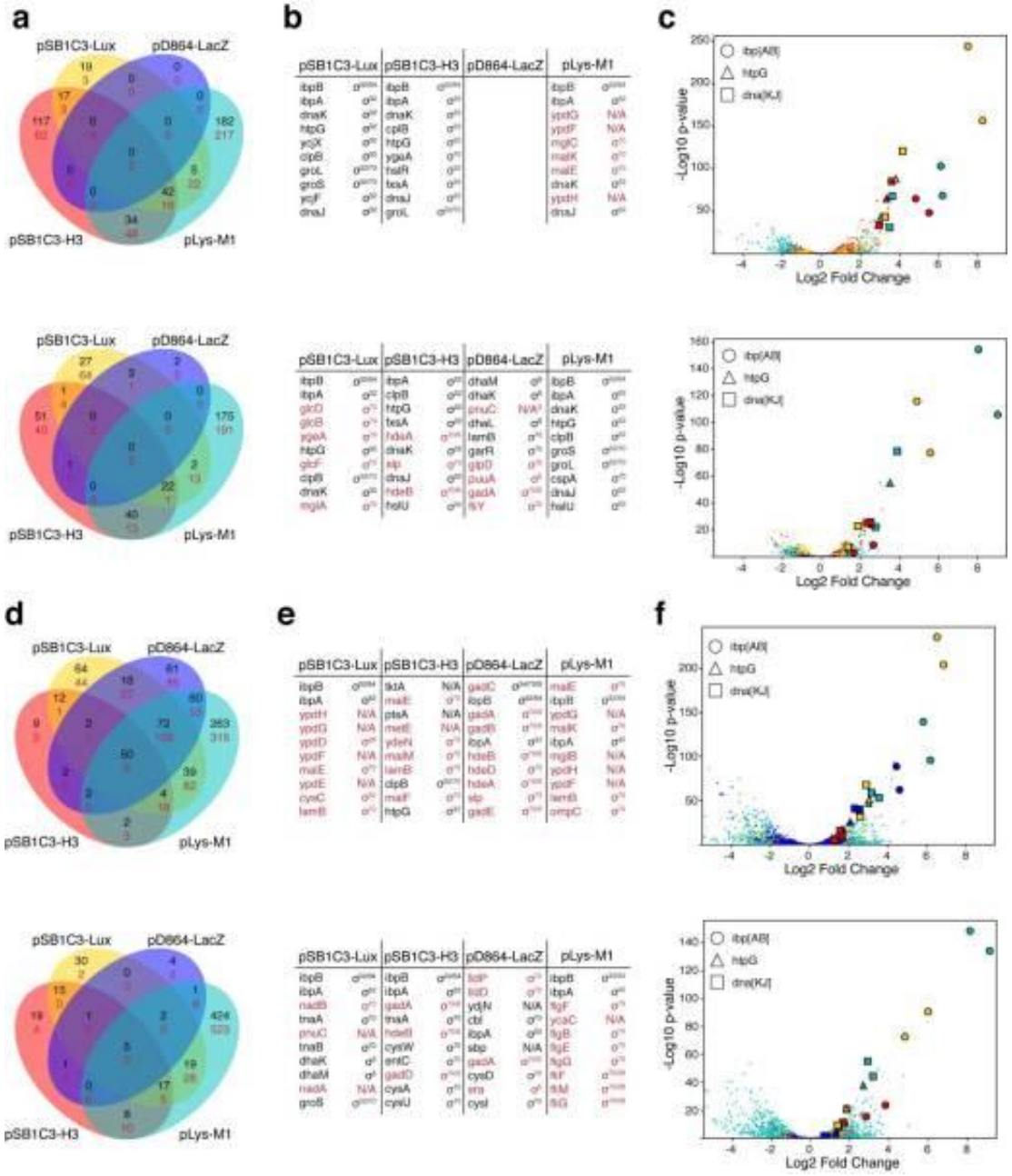
(a) Venn diagrams of differentially expressed genes in MG1655 *E. coli* 1h post-induction. The set of significantly differentially expressed native genes (alpha < 0.05) was identified separately for each of three synthetic gene circuits comparing cells transformed with the synthetic circuits to cells transformed with the corresponding empty plasmids. Up-regulated and down-regulated genes for each sample are displayed in black and red, respectively; **(b)** List of top ten differentially expressed genes (black are up-regulated, red are down-regulated). Tables report the top ten differentially expressed genes for each construct and the associated regulatory sigma factors. Bold names indicate genes differentially expressed also at 15 min post-induction. Left columns show the differentially expressed genes that are shared among all 3 constructs; **(c)** Volcano plots. Volcano plots show genes that display large changes and are also statistically significant (DeSeq2, n=3). Symbol colors correspond to the constructs colors displayed in (a) with pSB1C3-Lux, pD864-LacZ and pSB1C3-H3 displayed in yellow, blue and red, respectively. **(d)** Transcription profiles of the *dnaKJ*, *htpG*, *ibpAB and groSL* regulons and the GFP monitor cassette from DH10B cells with induced synthetic construct pD864-LacZ and pSB1C3-H3. Top profiles are for strains containing only the empty plasmid 15 min post-induction, middle and bottom profiles are for strains containing synthetic constructs 15 and 60 min post-induction, respectively. Solid grey regions show average and transparent red regions span the min to max of transcription profiles generated from three biological replicates. Genetic designs below each profile are drawn using SBOLv notation.



Supplementary Figure 5

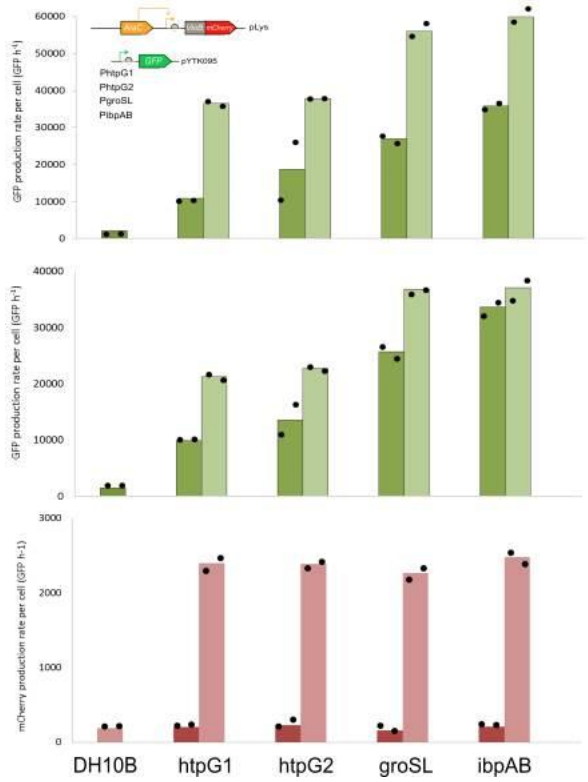
Global transcriptional changes in response to burden 15 min after induction of synthetic construct expression.

The set of significantly differentially expressed native genes (alpha < 0.05) was identified in DH10B (top panels) and MG1655 (bottom panels) cells separately for each of the three synthetic gene circuits pSB1C3-Lux, pSB1C3-H3 and pD864-LacZ, comparing cells transformed with the synthetic circuits to cells transformed with the corresponding empty plasmids. **(a)** Venn diagrams of differentially expressed genes. Venn diagrams report the number of up-regulated and down-regulated genes for each sample in DH10B and MG1655, which are displayed in black and red, respectively; **(b)** List of top ten differentially expressed genes (black are up-regulated, red are down-regulated). Tables report the top ten differentially expressed genes for each construct and the associated regulatory sigma factors. Left columns show the differentially expressed genes that are shared among all 3 constructs; **(c)** Volcano plots. Volcano plots show genes that display large changes and are also statistically significant (DeSeq2, n=3). Symbol colors correspond to the constructs colours displayed in (a) with pSB1C3-Lux, pD864-LacZ and pSB1C3-H3 displayed in yellow, blue and red, respectively.



Supplementary Figure 6

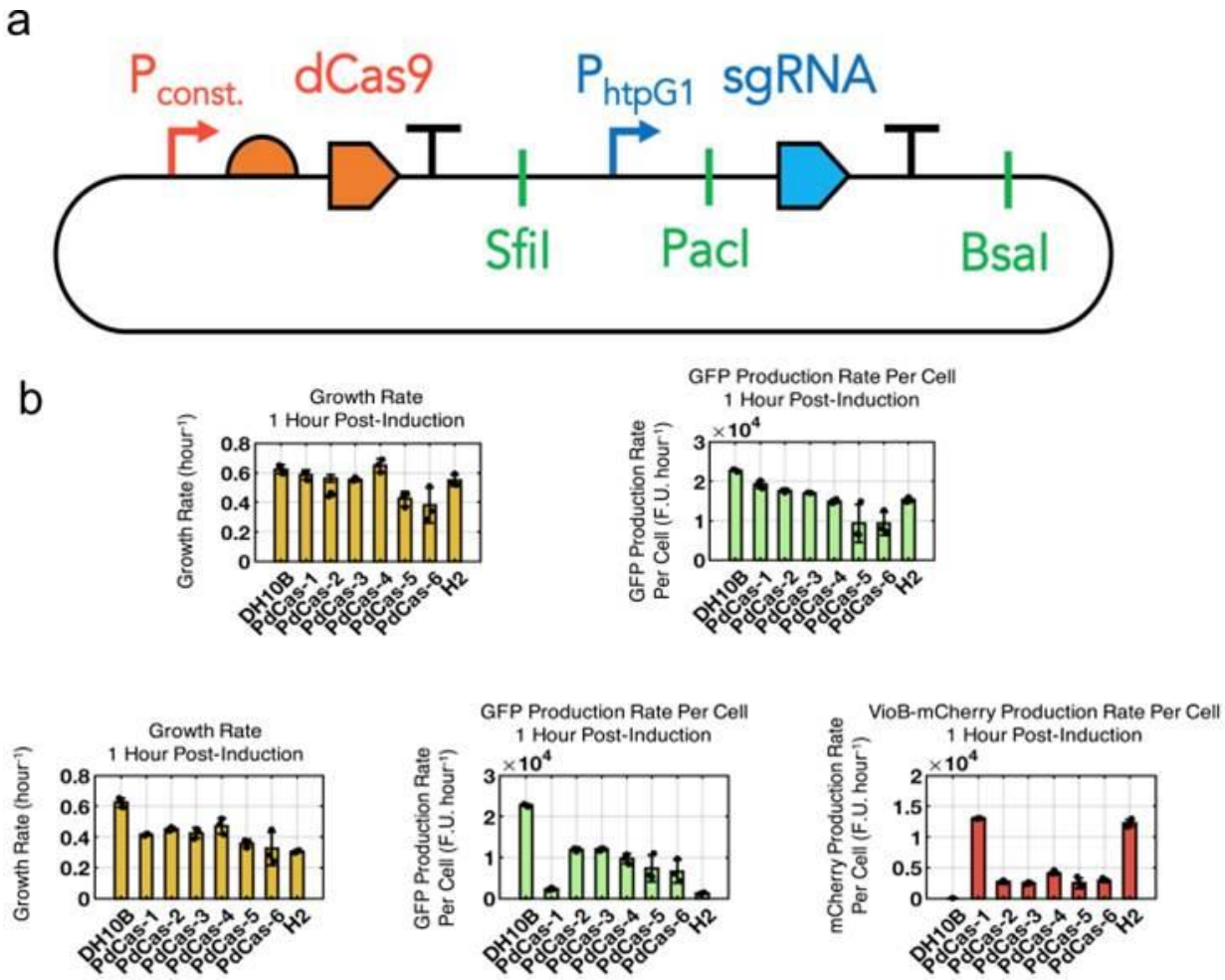
Global transcriptional changes in response to burden 15 and 60 min post-induction for all four synthetic constructs.
The set of significantly differentially expressed native genes (alpha < 0.05) at 15 min (panels a, b and c) and 60 min (panels d, e and f) after induction of gene expression was identified separately for each of four synthetic gene circuits pSB1C3-Lux, pSB1C3-H3, pLys-M1 and pD864-LacZ, comparing cells transformed with the synthetic circuits to cells transformed with the corresponding empty plasmids. **(a, d)** Venn diagrams of differentially expressed genes. Venn diagrams report the number of up-regulated and down-regulated genes for each sample in DH10B (top) and MG1655 (bottom), which are displayed in black and red, respectively; **(b, e)** List of top ten differentially expressed genes (black are up-regulated, red are down-regulated). Tables report the top ten differentially expressed genes for each construct and the associated regulatory sigma factors in DH10B (top) and MG1655 (bottom). **(c, f)** Volcano plots. Volcano plots show genes that display large changes and are also statistically significant (DeSeq2, n=3) in DH10B (top) and MG1655 (bottom). Symbol colours correspond to the constructs colours displayed in (a) with pSB1C3-Lux, pD864-LacZ, pSB1C3-H3 and pLys-M1 represented in yellow, blue, pink, and cyan respectively.



Supplementary Figure 7

Characterization of burden-dependent activation of plasmid-based σ^{32}-regulated promoters.

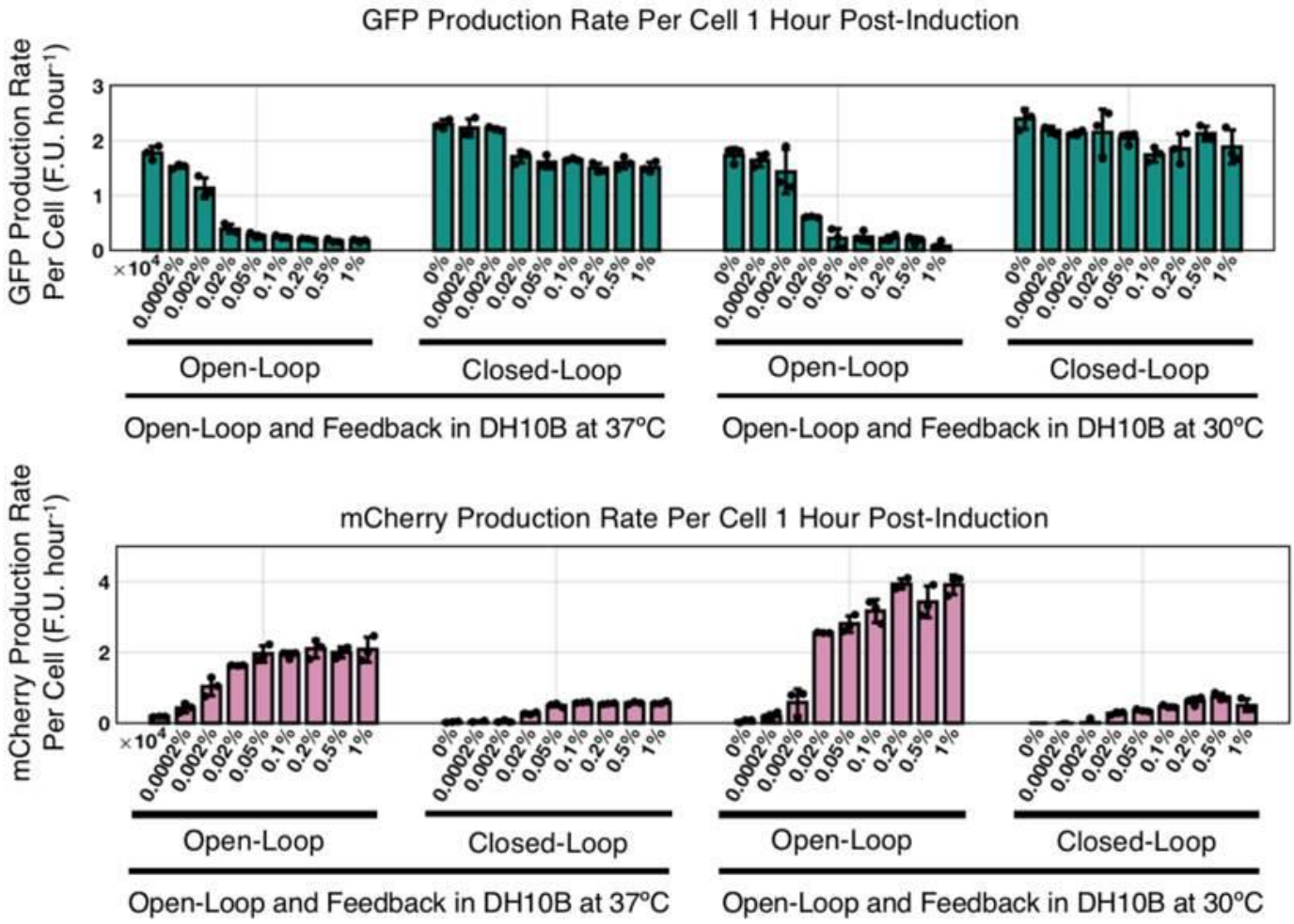
The *in vivo* response to burden from the *htpG1, htpG2, groSL* and *ibpAB* promoters. All four promoters were assessed for their ability to induce GFP expression from a high-copy plasmid when co-transformed with a construct expressing VioB-mCherry upon arabinose induction. Data show the GFP production rate per cell from the four promoters 2 hours (top panel) and 1 hour (middle panel) post-induction and VioB-mCherry expression 1 hour post induction (bottom panel) in DH10B cells. Dark and light green bars represent uninduced (no arabinose) and induced samples (arabinose), respectively. Dots show single measurements and bar show the mean value of two independent measurements.



Supplementary Figure 8

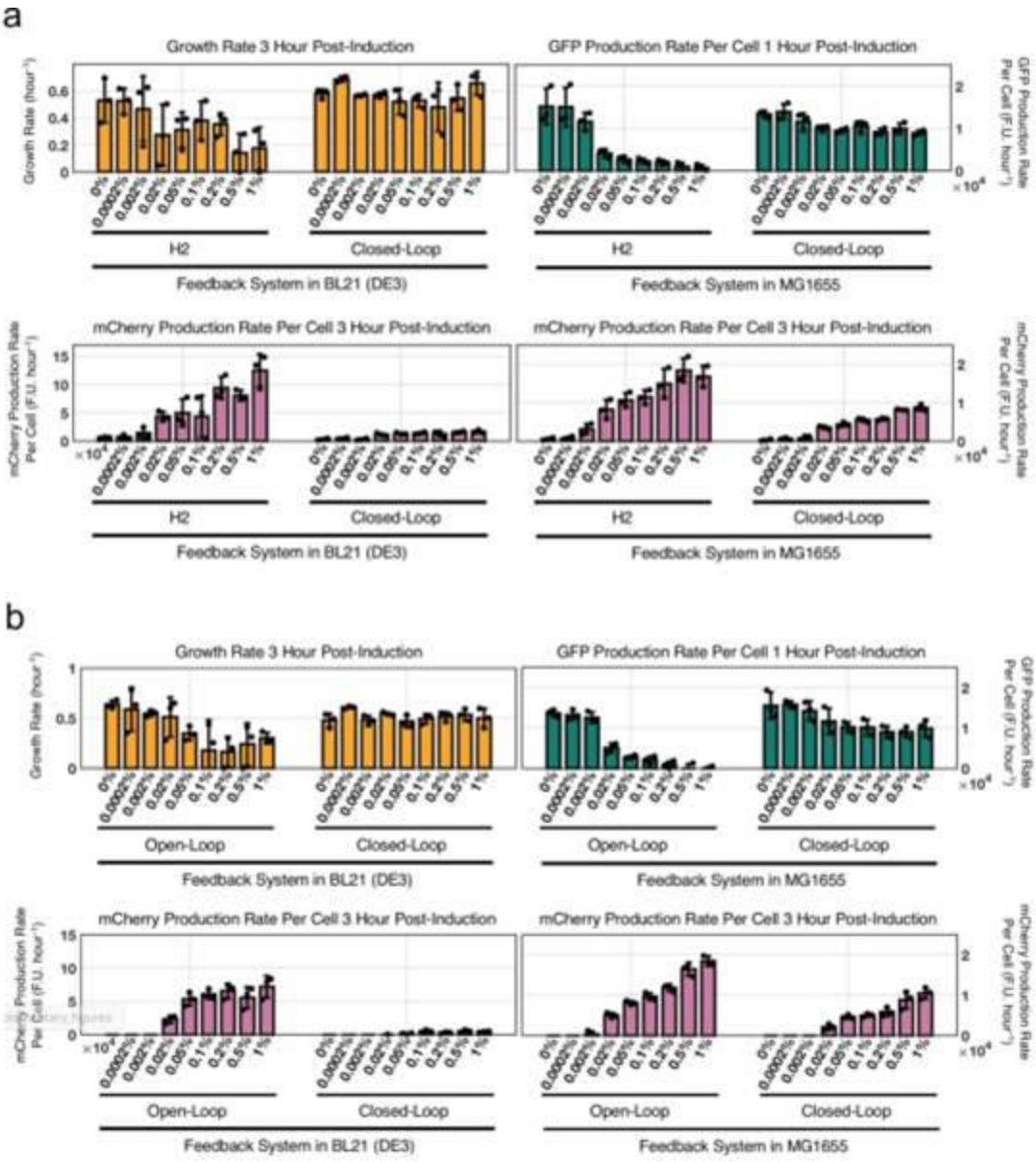
Biomolecular feedback controller.

(a) Modular structure of the feedback system. The feedback system was placed on a medium copy number plasmid with the ampicillin resistance marker. The feedback is composed of a dCas9 cassette with constitutive expression from a synthetic promoter and of a downstream burden-responsive cassette. In the burden-responsive cassette the burden-responsive htpG1 promoter controls the production of a sgRNA, in this case targeting the pBAD core promoter. The system was conceived with a modular structure where the htpG1 promoter and the gRNA can be easily replaced using rare restriction enzymes flanking each module. **(b)** Burden imposed by the molecular feedback controller on DH10B cells. Top panels show the growth rates (left panel) and GFP production rates (right panel) 1 hour post induction for uninduced DH10B cells (with integrated capacity monitor) when no construct is present (DH10B), and uninduced cells containing both pSB1C3-H2 and the feedback system plasmids (PdCas9-X), or uninduced cells containing just pSB1C3-H2 (H2). The burden imposed by the feedback system when promoters of different strength are present upstream of dCas9 is shown (PdCas9-1 to 6). Bottom panels show growth rates, as well as GFP and mCherry production rates when the same constructs are induced with arabinose. H2 is a previously-designed and tested construct, similar to pSB1C3-H3, but with a stronger RBS and so imposing even higher levels of burden. For a full description of H2, please see Ceroni *et al*, *Nature Methods* 2015. Dots show single measurements and bar show the mean value. Error bars show the standard deviation of three biologically independent repeats.



Supplementary Figure 9

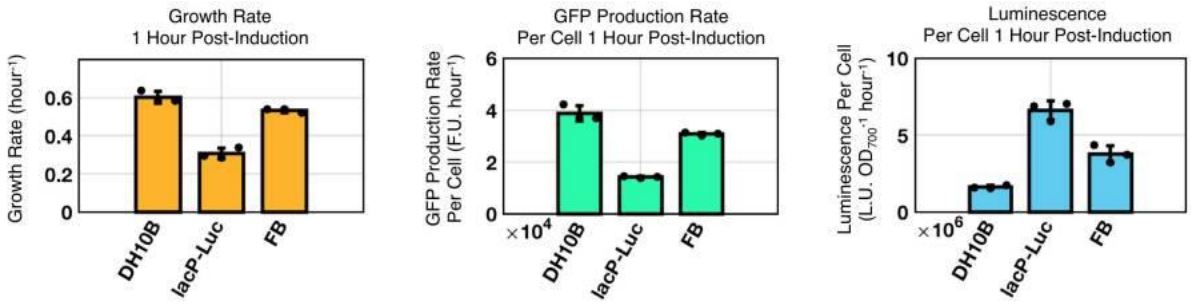
Test of the robustness of the feedback system at different temperatures.
GFP (top) and VioB-mCherry (bottom) production rate per cell are shown when different levels of arabinose are used to trigger VioB-mCherry production in DH10B cells at 37°C and 30°C. The performance of the feedback system in presence of construct pSB1C3-H2 (Closed-Loop) was compared to an 'Open-Loop' version where the sgRNA targeting pBAD had been replaced by a random sgRNA with no affinity for any promoter. For both strains VioB-mCherry output from pSB1C3-H2 (lower panels) and cellular capacity (top panels) were characterised when different levels of arabinose were used to induce construct expression. Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats.



Supplementary Figure 10

Portability of the feedback controller in multiple *E. coli* strains.

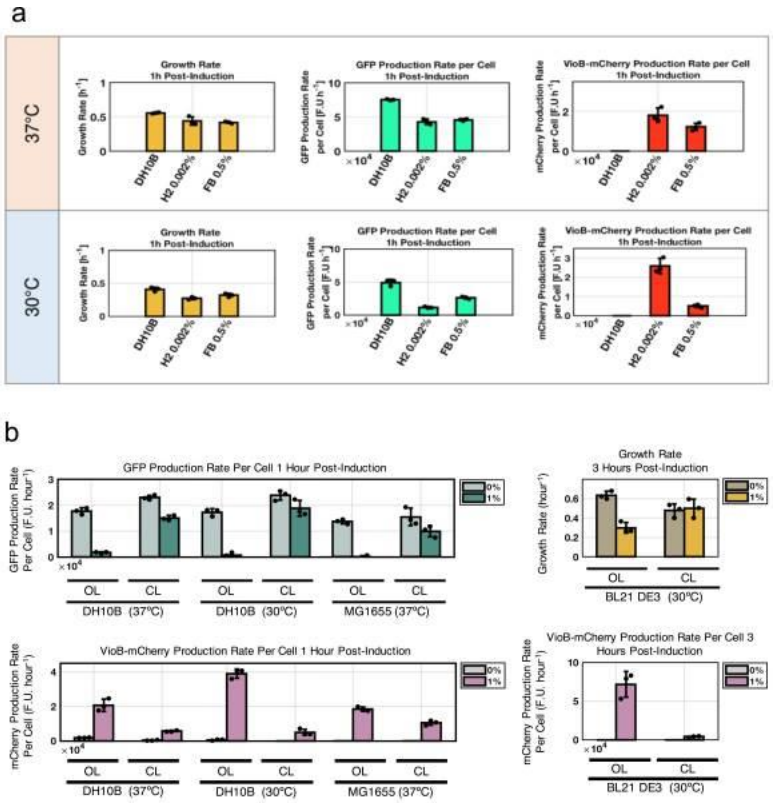
(a) Comparison between the feedback system and the unregulated strain. The functionality of the controller was tested in BL21-DE3 (left panels) and MG1655 (right panels). The performance of the feedback system in presence of construct pSB1C3-H2 (Closed-Loop) was compared to the strain where only pSB1C3-H2 is present (H2). For both strains VioB-mCherry output from pSB1C3-H2 (lower panels) and cellular capacity (top panels) were characterised when different levels of arabinose are used to induce construct expression. Since BL21-DE3 cells do not bear the capacity monitor, growth rate 3 hours post-induction was used as a proxy for cellular capacity instead of the GFP production rate per cell. Error bars show standard deviation of three independent repeats. **(b)** Comparison between the feedback system and the open loop strain. The performance of the feedback system in presence of construct pSB1C3-H2 (Closed-Loop) was compared to an 'Open-Loop' version where the sgRNA targeting pBAD had been replaced by a random sgRNA with no affinity for any promoter. Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats.



Supplementary Figure 11

The biomolecular feedback controller retargeted to a different synthetic construct.

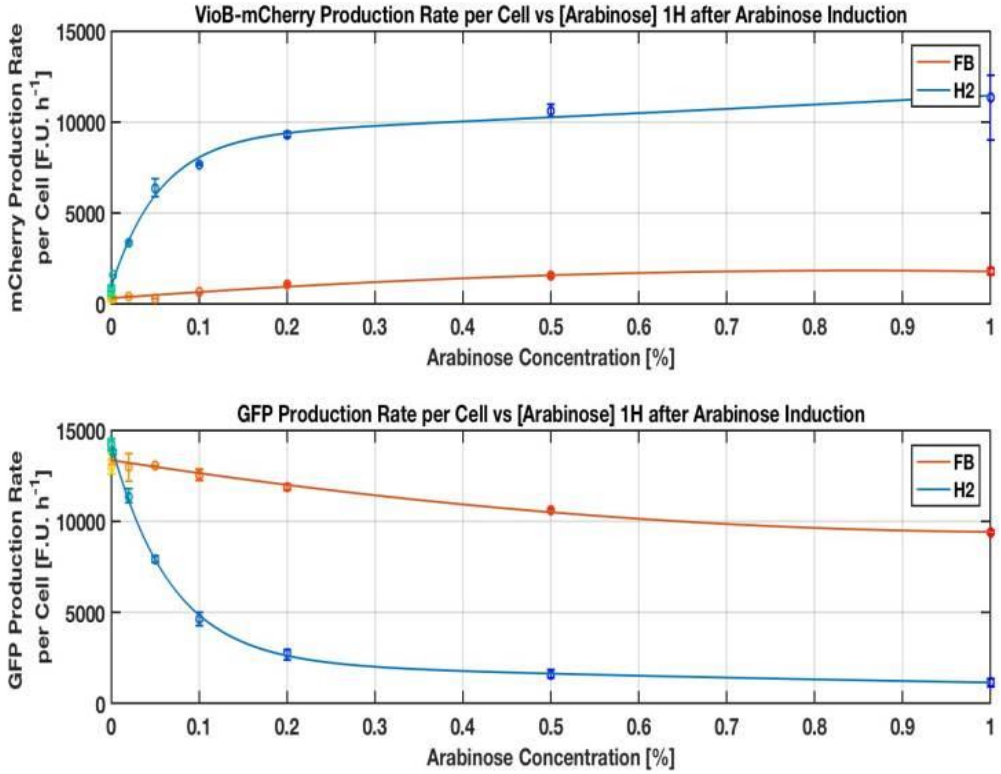
Growth rate (left), GFP expression rate (middle) and total luminescence per cell (right) are shown for a construct where the firefly luciferase operon is expressed under the control of the lacP promoter (lacP-Luc) in DH10B *E. coli*. In cells where the lacP-targeting closed-loop feedback controller is alongside the luciferase operon plasmid (FB), luminescence is decreased while growth rate and GFP production rate are improved. Sequences of the lacP core promoter and sgRNA targeting this are both shown in Table S7. Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats.



Supplementary Figure 12

Presence of the feedback controller confers robustness to temperature change.

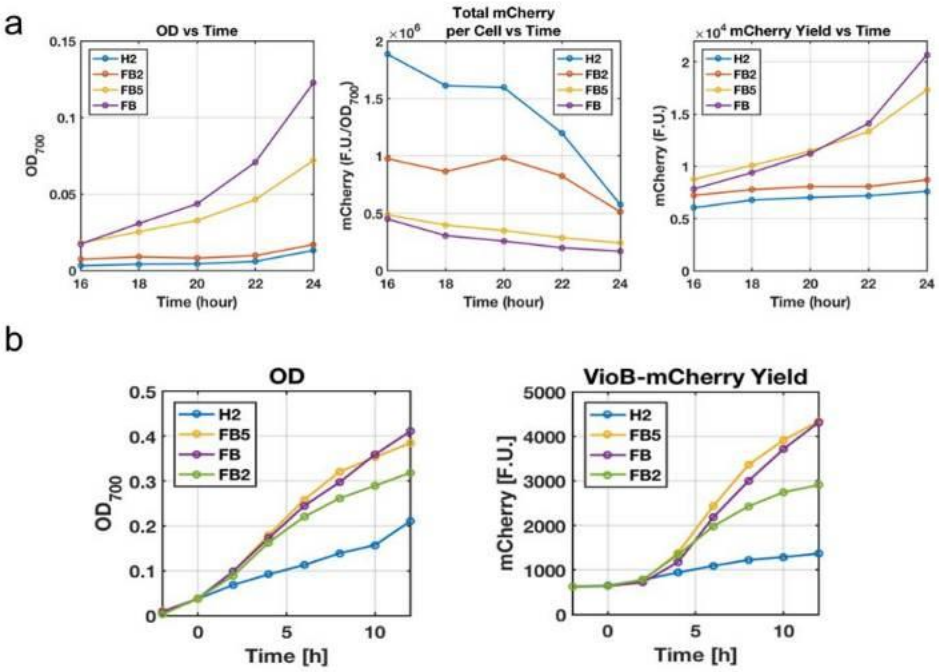
(a) Top panels show the growth rate, and GFP and VioB-mCherry production rates 1 hour post induction at 37ºC in DH10B cells. When pSB1C3-H2 with the closed-loop feedback control plasmid (FB) and pSB1C3-H2 alone (H2) strains are induced with 0.5% and 0.002% arabinose, respectively, they both behave very similarly 1 hour post-induction in terms of growth rate, and GFP and VioB-mCherry production rates. Lower panels show the growth rate, and GFP and VioB-mCherry production rates 1 hour post induction at 30ºC. At 30ºC, H2 now produces significantly more VioB-mCherry, but is accompanied by a decrease in cellular capacity and growth rate, while FB maintains VioB-mCherry expression and keeps the desired high expression capacity and growth. Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats. **(b)** The biomolecular feedback was tested in DH10B at 37°C and 30°C and MG1655 cells at 37°C with integrated capacity monitor (left panels) and in BL21-DE3 cells at 37°C (right panels). GFP (top) and VioB-mCherry (bottom) production rate per cell are shown for DH10B and MG1655 when 0% and 1% of arabinose are added. For BL21-DE3, growth rate (top) and VioB-mCherry rate (bottom) are shown instead. Dots show single repeats while bars show the mean value. Error bars show standard deviation of three independent repeats.



Supplementary Figure 13

Presence of the feedback controller confers robustness to varying concentrations of inducer.

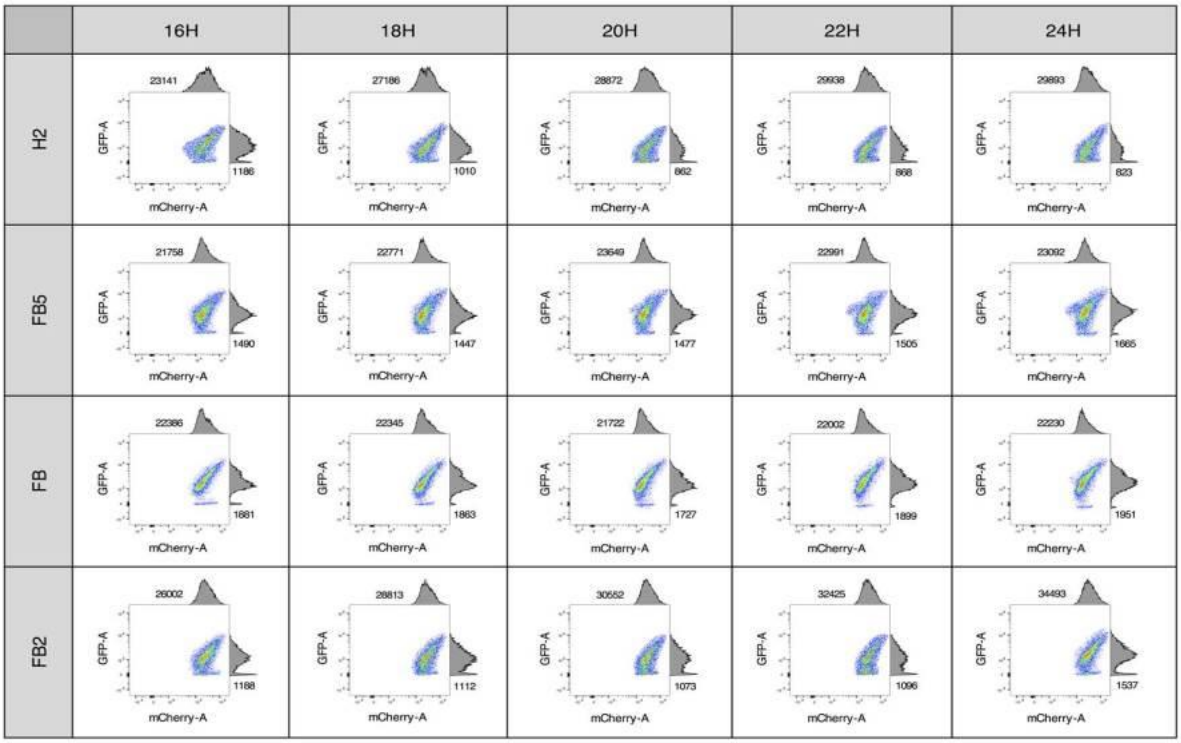
Top and bottom panels show VioB-mCherry and GFP production rates 1 hour post induction at different arabinose concentrations, respectively, in DH10B cells. While pSB1C3-H2 alone (H2) is highly sensitive to increasing concentrations of arabinose, pSB1C3-H2 with the feedback controller plasmid (FB) is capable of maintaining the GFP production rate per cell (i.e. the host capacity) and the VioB-mCherry production rates per cell within a much narrower range, thereby demonstrating robustness of the feedback system to inducer fluctuations. Dots represent the mean and the error bars indicate the maximum and minimum values of the triplicate.



Supplementary Figure 14

Burden-driven feedback control in a 24-h batch time course.

(a) Constructs FB, FB2 and FB5 from Figure 3c were selected for a 24 hour batch growth time course after addition of 1% arabinose to the cultures containing the feedback system and pSB1C3-H2 construct (H2). The culture density (left panel), total VioB-mCherry production per cell (middle panel) and total VioB-mCherry yield in growth flasks are compared for time points from 16 to 24 hours. Plots show how increased growth due to alleviation of burden can compensate the decreased VioB-mCherry production per cell, leading to higher total protein yields over time. Interestingly, looking at Figure S15 at 20 hours it is possible to compare the mCherry per cell values measured using the plate reader with the corresponding values obtained through flow cytometry measurement. Both plate reader and flow cytometry assays confirm H2 (no feedback control) and FB2 as the constructs producing the most VioB-mCherry per cell, while FB5 produces a lower amount of VioB-mCherry per cell, and FB (the closed-loop feedback controller) produces the least amount per cell. **(b)** Earlier time points of a 24 hour time-course experiment are shown. Data correspond to a time-course from 0 to 12 hours for culture density (left) and total VioB-mCherry yield (right) in growth flasks. To avoid artefacts due to measurements and normalisation with very low ODs, this experiment was started at a higher initial OD.



Supplementary Figure 15

Single-cell 24-h time-course experiment of the burden-driven feedback controls.

Graphs show single-cell flow cytometry measurements of the population corresponding to the plate reader data shown in Figure 3d. The geometric mean values of sfGFP and VioB-mCherry expression per cell are given next to the histogram distributions. Plots confirm that no escape mutant arose during the batch culture dynamic experiment. The experiment was repeated three times with similar results.

Supplementary Tables

Construct	Part	Sequence
pD864-LacZ	**pRHAM**	CACCACAATTCAGCAAATTGTGAACATCATCACGTTCATCTTT CCCTGGTTGCCAATGGCCCATTTTCCTGTCAGTAACGAGAA GGTCGCGAATTCAGGCGCTTTTTAGACTGG
	RBS	AGGAGGTAAAAA
	LacZ	ATGACCATGATTACGGATTCACTGGCCGTCGTTTTACAACGT CGTGACTGGGAAAACCCTGGCGTTACCCAACTTAATCGCCT TGCAGCACATCCCCCTTTCGCCAGCTGGCGTAATAGCGAAG AGGCCCGCACCGATCGCCCTTCCCAACAGTTGCGCAGCCT GAATGGCGAATGGCGCTTTGCCTGGTTTCCGGCACCAGAAG CGGTGCCGGAAAGCTGGCTGGAGTGCGATCTTCCTGAGGC CGATACTGTCGTCGTCCCCTCAAACTGGCAGATGCACGGTT ACGATGCGCCCATCTACACCAACGTGACCTATCCCATTACG GTCAATCCGCCGTTTGTTCCCACGGAGAATCCGACGGGTTG TTACTCGCTCACATTTAATGTTGATGAAAGCTGGCTACAGGA AGGCCAGACGCGAATTATTTTTGATGGCGTTAACTCGGCGTT TCATCTGTGGTGCAACGGGCGCTGGGTCGGTTACGGCCAG GACAGTCGTTTGCCGTCTGAATTTGACCTGAGCGCATTTTTA CGCGCCGGAGAAAACCGCCTCGCGGTGATGGTGCTGCGCT GGAGTGACGGCAGTTATCTGGAAGATCAGGATATGTGGCGG ATGAGCGGCATTTTCCGTGACGTCTCGTTGCTGCATAAACC GACTACACAAATCAGCGATTTCCATGTTGCCACTCGCTTTAA TGATGATTTCAGCCGCGCTGTACTGGAGGCTGAAGTTCAGA TGTGCGGCGAGTTGCGTGACTACCTACGGGTAACAGTTTCT TTATGGCAGGGTGAAACGCAGGTCGCCAGCGGCACCGCGC CTTTCGGCGGTGAAATTATCGATGAGCGTGGTGGTTATGCC GATCGCGTCACACTACGTCTGAACGTCGAAAACCCGAAACT GTGGAGCGCCGAAATCCCGAATCTCTATCGTGCGGTGGTTG AACTGCACACCGCCGACGGCACGCTGATTGAAGCAGAAGC CTGCGATGTCGGTTTCCGCGAGGTGCGGATTGAAAATGGTC TGCTGCTGCTGAACGGCAAGCCGTTGCTGATTCGAGGCGTT AACCGTCACGAGCATCATCCTCTGCATGGTCAGGTCATGGA TGAGCAGACGATGGTGCAGGATATCCTGCTGATGAAGCAGA ACAACTTTAACGCCGTGCGCTGTTCGCATTATCCGAACCATC CGCTGTGGTACACGCTGTGCGACCGCTACGGCCTGTATGTG GTGGATGAAGCCAATATTGAAACCCACGGCATGGTGCCAAT GAATCGTCTGACCGATGATCCGCGCTGGCTACCGGCGATGA GCGAACGCGTAACGCGAATGGTGCAGCGCGATCGTAATCAC CCGAGTGTGATCATCTGGTCGCTGGGGAATGAATCAGGCCA CGGCGCTAATCACGACGCGCTGTATCGCTGGATCAAATCTG TCGATCCTTCCCGCCCGGTGCAGTATGAAGGCGGCGGAGC CGACACCACGGCCACCGATATTATTTGCCCGATGTACGCGC GCGTGGATGAAGACCAGCCCTTCCCGGCTGTGCCGAAATG GTCCATCAAAAAATGGCTTTCGCTACCTGGAGAGACGCGCC CGCTGATCCTTTGCGAATACGCCCACGCGATGGGTAACAGT CTTGGCGGTTTCGCTAAATACTGGCAGGCGTTTCGTCAGTAT CCCCGTTTACAGGGCGGCTTCGTCTGGGACTGGGTGGATCA GTCGCTGATTAAATATGATGAAAACGGCAACCCGTGGTCGG

		CTTACGGCGGTGATTTTGGCGATACGCCGAACGATCGCCAG TTCTGTATGAACGGTCTGGTCTTTGCCGACCGCACGCCGCA TCCAGCGCTGACGGAAGCAAAACACCAGCAGCAGTTTTTCC AGTTCCGTTTATCCGGGCAAACCATCGAAGTGACCAGCGAA TACCTGTTCCGTCATAGCGATAACGAGCTCCTGCACTGGAT GGTGGCGCTGGATGGTAAGCCGCTGGCAAGCGGTGAAGTG CCTCTGGATGTCGCTCCACAAGGTAAACAGTTGATTGAACTG CCTGAACTACCGCAGCCGGAGAGCGCCGGGCAACTCTGGC TCACAGTACGCGTAGTGCAACCGAACGCGACCGCATGGTCA GAAGCCGGGCACATCAGCGCCTGGCAGCAGTGGCGTCTGG CGGAAAACCTCAGTGTGACGCTCCCCGCCGCGTCCCACGC CATCCCGCATCTGACCACCAGCGAAATGGATTTTTGCATCGA GCTGGGTAATAAGCGTTGGCAATTTAACCGCCAGTCAGGCT TTCTTTCACAGATGTGGATTGGCGATAAAAAACAACTGCTGA CGCCGCTGCGCGATCAGTTCACCCGTGCACCGCTGGATAAC GACATTGGCGTAAGTGAAGCGACCCGCATTGACCCTAACGC CTGGGTCGAACGCTGGAAGGCGGCGGGCCATTACCAGGCC GAAGCAGCGTTGTTGCAGTGCACGGCAGATACACTTGCTGA TGCGGTGCTGATTACGACCGCTCACGCGTGGCAGCATCAG GGGAAAACCTTATTTATCAGCCGGAAAACCTACCGGATTGAT GGTAGTGGTCAAATGGCGATTACCGTTGATGTTGAAGTGGC GAGCGATACACCGCATCCGGCGCGGATTGGCCTGAACTGC CAGCTGGCGCAGGTAGCAGAGCGGGTAAACTGGCTCGGAT TAGGGCCGCAAGAAAACTATCCCGACCGCCTTACTGCCGCC TGTTTTGACCGCTGGGATCTGCCATTGTCAGACATGTATACC CCGTACGTCTTCCCGAGCGAAAACGGTCTGCGCTGCGGGA CGCGCGAATTGAATTATGGCCCACACCAGTGGCGCGGCGA CTTCCAGTTCAACATCAGCCGCTACAGTCAACAGCAACTGAT GGAAACCAGCCATCGCCATCTGCTGCACGCGGAAGAAGGC ACATGGCTGAATATCGACGGTTTCCATATGGGGATTGGTGG CGACGACTCCTGGAGCCCGTCAGTATCGGCGGAATTCCAGC TGAGCGCCGGTCGCTACCATTACCAGTTGGTCTGGTGTCAA AAATAA
pSB1C3-H3		see Ceroni et al[1]
pSB1C3-Lux		see Ceroni et al[1]
pLys-M1		see Ceroni et al[1]
pSB1C3-H2		see Ceroni et al[1]

Supplementary Table 1. Sequences of the parts composing the four constructs used in this study. For pD864-LacZ sequences of the pRhaBAD promoter, RBS and LacZ coding gene are reported; for pSB1C3-H3, pSB1C3-Lux and pLys-M1 information can be found in our previous publication[1].

Strain	Plasmid	Synthetic circuit
DH10B	pSB1C3	-
DH10B	pSB1C3	H3
DH10B	pLys	M1
DH10B	pLys	-
DH10B	pSB1C3	Lux
DH10B	pD864	-
DH10B	pD864	LacZ
DH10B	-	-
MG1655	pSB1C3	-
MG1655	pSB1C3	H3
MG1655	pLys	M1
MG1655	pLys	-
MG1655	pSB1C3	Lux
MG1655	pD864	-
MG1655	pD864	LacZ

Supplementary Table 2. Synthetic constructs selected for the present study. Four constructs plus controls were used for the *in vivo* assay and RNAseq. Three replicates were repeated independently on three different days for a total of ninety samples shaping the final data set (7 constructs X 2 strains X 3 replicates X 2 time points = 84 samples; plus DH10B:GFP cells X 3 replicates X 2 time steps)

	15 minutes		60 minutes	
	mean	**std**	**Mean**	**std**
pD864	1.2%	0.2%	1.5%	0.0%
pD864-LacZ	11.7%	1.2%	22.9%	2.2%
pSB1C3	0.6%	0.0%	0.7%	0.1%
pSB1C3-H3	39.4%	5.8%	33.3%	4.9%
pSB1C3-Lux	38.1%	4.1%	42.7%	4.9%
pLys	11.6%	1.5%	12.8%	0.7%
pLys-M1	47.3%	12.7%	52.9%	2.7%

Supplementary Table 3. Impact of synthetic expression on global transcription of DH10B cells. Raw RNAseq reads were used to calculate the impact of gene expression from the synthetic constructs on the global transcription of the cells in DH10B at 15 and 60 minutes after induction. Values are reported as percentage and were calculated as follows: *% = (sum of all plasmid-mapped read numbers) / (sum of all raw mapped read numbers).* Table shows average and standard deviation of three independent repeats.

	15 minutes		60 minutes	
	mean	**std**	**Mean**	**std**
pD864	1%	0.2%	1%	0.2%
pD864-LacZ	7.9%	0.7%	21.5%	1.3%
pSB1C3	1.1%	0.2%	0.8%	0.1%
pSB1C3-H3	33.8%	2.3%	30.5%	0.8%
pSB1C3-Lux	14.7%	1.3%	20.7%	2.9%
pLys	4.4%	0.5%	3.4%	0.3%
pLys-M1	23.3%	4.7%	35.6%	3.8%

Supplementary Table 4. Global impact of synthetic expression on transcription of MG1655 cells. Raw RNAseq reads were used to calculate the impact of gene expression from the synthetic constructs on the global transcription of the cells in MG1655 at 15 and 60 minutes after induction. Values are reported as percentage and were calculated as follows: *% = (sum of all plasmid-mapped read numbers) / (sum of all raw mapped read numbers)*. Table shows average and standard deviation of three independent repeats.

htpG1	**GGCCGCCTAGGCC**acgagagcaggatcacctgctctcgcttgaaattattctcccttgtccc catctctccca**TTAATTAA**
htpG2	**GGCCGCCTAGGCC**ggatcacctgctctcgcttgaaattattctcccttgtccccatctctccca catcctgtt**TTAATTAA**
groSL	**GGCCGCCTAGGCC**cagaatttttttctttttcccccttgaaggggcgaagcctcatccccattt ctctggtc**TTAATTAA**
ibpAB	**GGCCGCCTAGGCC**tgcaaaaaaaagtccgctgataaggcttgaaaagttcatttccagac ccatttttacatc**TTAATTAA**

Supplementary Table 5. Sequences of the σ32-regulated promoters used in this study. The *htpG1*, *htpG2*, *groSL* and *ibpAB* promoters from *E. coli* were ordered as gBlocks from IDT. Each promoter has a SfiI and PacI restriction site upstream and downstream its sequence, respectively. Restriction sites are shown in bold while promoter sequences are shown in normal black.

Part	Sequence
dCas9 constitutive promoter	TTTACGGCTAGCTCAGTCCTAGGTATTGTGCTAGC
sgRNA in FB	**GAGTTGCGATAAAAAGCGTC**
sgRNA in FB1	GAGTTGCGAT-AAAAGCGTG
sgRNA in FB3	GAGTTGCGATAAAAAGCGTG
sgRNA in FB4	GAGTTGCGATAAAAAGGGTC
sgRNA in FB5	GAGTTGCGATAAAAGGCGTC
sgRNA in Open Loop FB	GCACAGATTTCAGGACTTAT

Supplementary Table 6. Sequences of the constitutive promoter guiding expression of dCas9 in the feedback device and of the library of sgRNAs used to implement the burden-based feedback system and its open loop counterpart. Top row shows the sequence of the constitutive promoter guiding the expression of dCas9 in the feedback device. The table also reports the library of point-mutated sgRNA developed to obtain a library of burden-responsive feedbacks that result in different construct outputs and corresponding cellular capacity. Point mutations are shown in red while the original sgRNA sequence is shown at the top [sgRNA in FB]. The sequence of the control sgRNA used for construction of the Open Loop feedback is shown in the final row.

Part	Sequence
LacP core promoter	TTTACACTTTATGCTTCCGGCTCGTATAATGTGTGGA
Luc operon plasmid	see Ceroni et al[1]
gRNA targeting LacP	CTTCCGGCTCGTATAATGTG

Supplementary Table 7. Sequences of the parts composing the Firefly Luc construct and the sgRNA targeting the core lacP promoter. Top row shows the sequence of the constitutive LacP core promoter guiding the expression of the Luc operon in the Luc construct used to test the feedback controller for portability. Bottom row shows the sequence of the sgRNA targeting the lacP core promoter in the Luc construct.

Supplementary Notes

Supplementary Note 1

Selection of inducible-expression constructs chosen to trigger burden, and description of their genetic content.

We chose a set of four exemplar inducible synthetic constructs with different genes, codon-optimisation and regulation. These were focused on three different exemplar cases of plasmid-based inducible expression; (i) inducible reporter expression, (ii) overexpression of a large non-functional heterologous protein, and (iii) expression of an operon encoding a metabolic pathway operon**.**

For (i), the inducible reporter, we used plasmid **pD864-LacZ**, which expresses a copy of the *E. coli* LacZ enzyme from a rhamnose-inducible promoter. This construct was selected as it only contains genes and promoter sequences native to *E. coli* (*i.e*. the pRhaBAD promoter and the LacZ open reading frame) taken out of their native context. The backbone plasmid used (**pD864**) has a standard ampicillin selectable marker.

For (ii), the large non-functional heterologous protein, we used plasmid **pSB1C3-H3** which expresses the enzyme VioB with a C-terminal mCherry fusion from an arabinose-inducible promoter. This construct was used in our previous study and is known to produce a burden simply from the act of expressing VioB which does not play any role in the cell [1]. The open reading frame is codon-optimized for *E. coli* and the plasmid contains a version of the commonly-used araBAD induction cassette (AraC-pBAD) with a point mutation that gives slightly higher expression than the standard araBAD induction cassette. The backbone plasmid (**pSB1C3**) is a high-copy plasmid with chloramphenicol selectable marker.

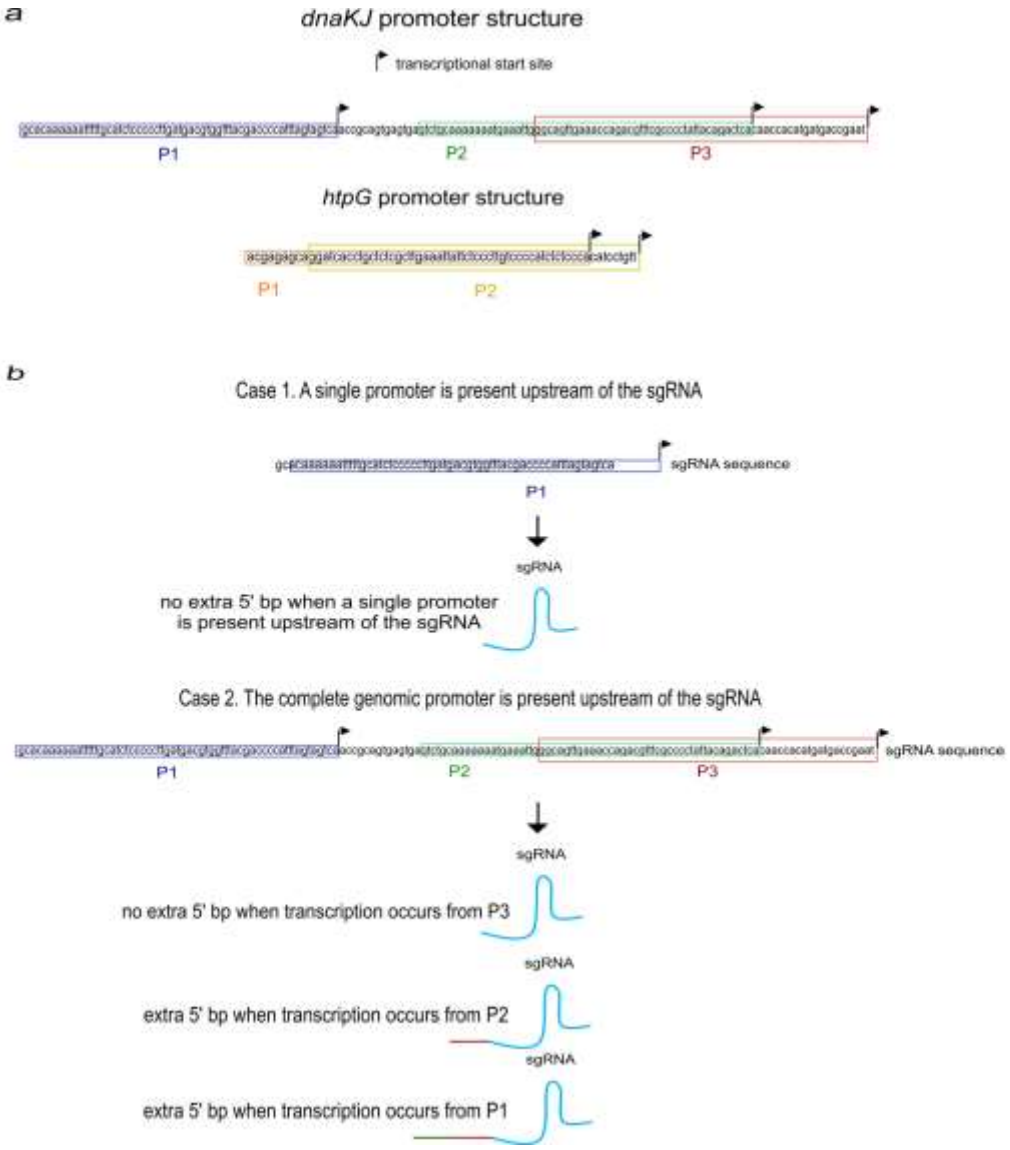
For (iii), the metabolic pathway operon, we used plasmid **pSB1C3-Lux** which expresses the multigene Lux operon from *Vibrio fischeri* (codon-optimised for *E. coli*) from the standard araBAD induction cassette (AraC-pBAD). This also has the **pSB1C3** plasmid backbone (high-copy, with chloramphenicol selectable marker).

Finally, for preliminary investigations we also added a fourth synthetic construct called **pLys-M1**. This is an alternative design to pSB1C3-H3 that was used in our previous work and is known to impose more burden despite being on a lower-copy plasmid, due to having a stronger ribosome binding site sequence (RBS) [1]. It uses the **pLys** medium-copy plasmid backbone with chloramphenicol selectable marker.

Supplementary Note 2

The *dnaKJ* promoter as a candidate burden biosensor for use in the design of a biomolecular feedback controller

The promoter driving genomic expression of the *dnaKJ* regulon was one of the four promoters that showed high up-regulation in response to burden induction in our RNAseq analysis, together with the *ibpAB, groSL* and the *htpG* promoters. With the idea of utilising these promoters as biosensors for burden and implementing them in the design of a biomolecular feedback controller, we decided to assess whether their burden-based induction could be recreated when taken out of their natural genomic context and used to express GFP from a plasmid. This proved to be complicated for the *htpG* and *dnaKJ* regulons as these are transcribed from two and three overlapping promoters, respectively (Panel a below) [2]. For both promoters we ordered synthesised DNA encoding the five individual promoters (*htpG1, htpG2, dnaK1, dnaK2 and dnaK3*) so that their burden-responsive behaviour could be verified. Upon cloning upstream of a GFP cassette on a plasmid, *htpG1* and *htpG2* were able to act as promoters and were both induced by host burden. However, neither of the three *dnaK* constructs cloned were able to show appropriate promoter behaviour. Indeed, only the complete promoter as it appears in the genome was able to any induce GFP expression upon burden induction. While this would normally suffice as a burden-responsive promoter, we did not want to use this in our feedback system as the presence of three overlapping promoters would mean the possibility of transcribing RNA from 3 different transcription initiation sites. As our feedback controller requires burden-responsive sgRNA expression it is important that we can be assured of transcription from a single known transcription initiation site so that the expressed sgRNA maintains the same sequence, structure and functionality in all cases (Panel b below). For this reason the *dnaK* promoter was not explored further and the analysis was carried on considering only the *ibpAB*, *groSL*, *htpG1* and *htpG2* promoters.



Genomic structure of the *htpG* and *dnaKJ* promoters. (a) The promoters upstream of the dnaKJ and htpG genes comprise three (top) and two (bottom) different possible promoters, respectively. **(b)** When the separate promoters are cloned upstream of the sgRNA sequence, a unique transcriptional initiation site is present and thus one sgRNA is transcribed (Case 1). However, when single promoters cannot be used, the complete genomic promoter sequence generates multiple transcriptional initiation sites leading to extra bases at the 5' of the sgRNA, which could impede its correct functionality.

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Supplementary Note 3

Depletion of transcriptional and translational resources contributing to burden

A major finding in this study was that one hour post-induction, between 20% and 50% of all mRNAs in our RNAseq samples mapped to the genes from the synthetic constructs. As mRNA production in growing *E. coli* is estimated to represent approximately half of all active transcription[3] this means that our high burden constructs were likely using up as much as a quarter of all transcriptional resources when expressed. This suggests that limitations on transcription may be playing a larger role in expression burden than previously appreciated. Indeed, given that it is well-established that translation is the major resource cost in gene expression, we anticipated that a lack of translational resources, such as free ribosomes, would predominantly explain the decreased GFP expression rates we see from our capacity monitor gene in burdened cells. However, in this study we were able go further and better determine the different contributions of transcription and translation to this observed decrease by comparing our GFP expression rates to the corresponding RNAseq quantification for the GFP-encoding transcript (although it must be stressed that the GFP measurement is a rate, while the RNA levels are a relative amount, and so the two are not directly comparable).

The surprising observation from our data was that the relative levels of the GFP mRNAs per cell (deduced from RNAseq FPKM values) closely matched the equivalent GFP expression rates per cell from the *in vivo* assay (Supplementary Figure 2). This suggests that the decreased GFP production we observe stems from a decrease in the transcriptional activity of the gene, and not from changes in translational activity. This seems contradictory to the widely-held view that translational resource limitations predominately explain gene expression burden. However, it must be remembered that the effects on transcription also directly impact translation, especially if the mRNAs expressed from synthetic constructs have strong RBS sequences which will quickly sequester ribosomes, chaperones and other translational resources[4].

To fully understand the relationship between transcription and translation in burden will require capturing further data in future studies that bring in new experiments. RNAseq with spike-in controls standards[5] can quantify real mRNA numbers per cell and modifications to RNAseq now also enable mRNA degradation rates to be measured[6]. RiboSeq (an adaptation to RNAseq protocols that determines the footprints of all ribosomes on mRNAs in cells[23]) would also be especially valuable, allowing the rate of translation of the mRNAs to be inferred from the positions/densities of ribosomes on the transcripts in cells. Indeed, in recent work by Arkin and Cambray, all of these methods are used to assess how gene expression efficiency and *E.coli* growth rates are affected by 244,000 different designs of mRNAs that encode GFP [6]. An important point that arises from their work is that decreased translation of highly-expressed genes (e.g. our GFP gene with strong promoter, strong RBS and codon-optimized sequence) leads to increased degradation of its mRNAs by native RNAses. Thus, when the translation rate of such an mRNA decreases, the mRNA level itself decreases. Based on this knowledge, we attribute the decreased GFP mRNA levels in our burdened samples to both reduced transcription (directly) and reduced translation (indirectly). The two processes are clearly closely-intertwined and both are equally likely to

be affected by overexpression of a heterologous gene, as the resources needed for them (polymerase, sigma factors, ribosomes, tRNAs, chaperones, etc.) are unlikely to be in excess in fast-growing cells. However, despite both transcription and translation being affected and likely limiting, we still believe that the impact of burden (i.e. decreased growth rate) is predominantly due to the strain on translational resources, as these are an order of magnitude more costly for the cell to produce compared to the resources needed for transcription [7].

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